

Dear Hiring Committee,

Thank you for considering me for the Research Associate in Microbiome Ecology position. My understanding is that this is a highly interdisciplinary position. With the following, I aim to highlight the most relevant skills and experience to the position, by addressing the person specification criteria in grouped sections.

Person Specification

Essential Skills, Knowledge and Experience

Have, or be about to obtain, a PhD in a relevant field (e.g. ecology, evolutionary biology, microbiology, mathematical biology etc)

My PhD focuses on microbial ecology and challenges derived from amplicon and shotgun metagenomics sequencing data and their analysis. In the first year, I continued my MSc thesis project to deliver PEMA [1] an amplicon analysis workflow for 5 different marker genes. Then, I served as a lead developer of the PREGO data integration platform, designed to extract and consolidate associations among species, environments, and biological processes from literature and abundance tables from key platforms such as MGnify, and IMG/JGI [2]. PEMA has been integrated and supports the LifeWatch-ERIC Internal Joint Initiative on Autonomous Reef Monitoring Structures (ARMS) (<https://www.lifewatch.eu/internal-joint-initiative/workflows/>).

More than developing tools, I also had the chance to apply them to real world data. Along with Dr. Christina Pavloudi, we studied a hypersaline microbial swamp in Karpathos, Greece, applying both amplicon and shotgun sequencing. Combining our expertises, we managed to reconstruct a set of 250 bacterial and 39 archaeal metagenome assembled genomes (MAGs) revealing an essential diversity of potentially novel phylogenetic clades. Both sequencing methods gave us similar taxonomic compositions and most interestingly, revealed that the formation of the microbial mat in this marsh exhibits seasonal variation. On top, we found that the *salt-in* strategy was the most common for osmoadaptation of the taxa, and we ended up with a new hypothesis on the role of the anaplerotic reactions in which our samples were rather enriched with, and their role in replenishing TCA cycle intermediates, and thus on growth in such conditions.

Further work on amplicon analysis challenges, led me to develop a COI-based phylogenetic tree of life, which I then used to distinguish non-eukaryotic COI sequences from eukaryotic ones [3]. This approach was recently used to the 4th version of Crest, a widely used software for amplicon taxonomic classification. Also, in the last year of my PhD, and in collaboration with the MGnify team of EMBL-EBI, I was the leading developer of *metagoflow* [4] a MGnify-based pipeline, tailored to the needs of the European Marine Biological Research Centre (EMBRC) research infrastructure.

Last but not least, during the second year of my PhD, I joined the GeomScale project, a group of computational geometrists interested in applying their methods in real-world problems. Since then, I started studying metabolic models, the challenges to build them and the various ways for their analysis. Along with Dr. Apostolos Chalkis, we managed to establish a Multiphase Monte Carlo Sampling (MMCS) algorithm for efficient flux sampling [5] and sample for the first time on the solution space of the Recon3D model; the latest version of the human Genome Scale Metabolic model (GEM). As demonstrated below, metabolic modeling has become my primary tool for investigating my core research interest: microbial interactions.

Strong conceptual understanding of eco-evolutionary theory, particularly as applied to microbial communities A keen interest in developing new eco-evolutionary theory

In my first post-doc, I have been working on the 3D'omics project aiming at developing a 3D-based framework to disentangle microbe-microbe and animal-microbe interactions. Over this time, I focused on microbial interactions using metabolic modeling based approaches.

In a recent study of ours, we were able to show that predicted growth rates and their ratios (i.e., the predicted interaction strength between two taxa), do not correlate with *in vitro* data [6]; however, curated GEMs had more reliable results. To address this limitation, we have just released the first beta-version of a microbial growth data repository, called μ growthDB (<https://mgrowthdb.gbiomed.kuleuven.be/>). The broader the endorsement by the microbiology community, the more effectively it will enable us to constrain models with higher accuracy.

My main focus though was *microbetag* (<https://microbetag.readthedocs.io/>) a software ecosystem for annotating co-occurrence microbial networks. *microbetag* annotates taxa (nodes) with phenotypic traits, based on genomic and literature information, and their predicted associations (edges) with potential metabolic interactions, based on the *seed* concept. We consider seeds the compounds that are present and can be used from a reaction in a GEM, but no internal reaction is producing them, thus the species would have to get them from the environment to enable the corresponding reaction(s). Combining such annotations with network clustering algorithms and enrichment analysis, we are now able to produce specific hypotheses on what may drive a system. Only last week, *microbetag* was accepted in principle for publication in *Genome Biology* [7].

However, it is my belief, that what depicts my understanding on how microbial communities are shaped and evolved, but also my personal scientific motivation, is my proposal for the European Molecular Biology Organization (EMBO) postdoctoral fellowship program, which made it to the interviews stage. My main hypothesis was that the auxotrophies a species develops are related to its resource hierarchy—that is, its nutrient preferences. Moreover, since species in natural communities never exist in isolation, metabolic interactions with other community members would also influence auxotrophies, while resource hierarchies simultaneously shape community structure in a feedback loop. Working

further this framework, I believe will lead to great insight on the mechanisms communities are shaped.

Experience designing and analysing mathematical models of biological systems Experience testing models with data

In our perspective paper on metabolic modeling on human gut microbiome [8] we highlight how one can a set of different modeling approaches using a toy model of three common gut species with different metabolic capabilities: a carbohydrate fermenter, an acetogen, and a butyrate producer. We display how one can use kinetic models, inability to handle flexible metabolic responses means that in many cases it does not meet Einstein's famous 'as simple as possible but not simpler' criterium, and thus, we argue on the added value metabolic modeling can support.

In the following sections I provide more details on my previous projects, see for example our evaluation study for FBA-based interaction strength predictions [6]. Yet here, I would like to point out my ongoing project on *Salmonella infantis* and a set of probiotics known to decrease its abundance. To decipher how this is happening, whether all probiotics are required for the decrease to occur and whether the interactions among any of these strains are metabolically-driven, I am working with a set of *in vitro* data from mono- and co-culture communities, and the GEMs of the corresponding taxa.

A manuscript for this project is in preparation.

Ability to present in both written and oral publications

In the following, I highlight my most significant manuscripts and talks, focusing on those most relevant to the Research Associate in Microbiome Ecology position.

Strong journal publication record. Track record of peer-reviewed scientific publication(s)

To date, I have co-authored 17 peer-reviewed publications, serving as first author on 9 and corresponding author on 4. My PhD related work, was mostly published in the Giga-Science journal: both PEMA [1] and metagoflow [4] as well as the review I wrote on the IMBBC High Performance Computing (HPC) system [9]. Our study with Dr. Pavloudi on the hypersaline swamp in Karpathos, was published in FEMS Microbiology Ecology [10]. The MMCS algorithm for efficient flux sampling, was published in the Symposium on Computational Geometry [5], while the Python library implementing MMCS and a great range of other walks (*dingo*), was published in Bioinformatics Advances [11]. The evaluation on microbial interaction prediction using FBA-based analysis was published in BMC Bioinformatics [6]. Our gap-filler, *dnngior* was recently published in iScience [12] while *microbetag* was accepted in Genome Biology. For a complete list of my publications, see my CV.

Track record of conference and / or workshop oral and / or poster presentation(s)

I consider my self lucky to have joined in a great number of conferences and workshops, even if the Sars-Cov-2 pandemic was during my PhD. At Mikrobiokosmos 2022, the

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annual conference of the Greek microbiology community, my presentation on flux sampling was recognized with a talk award. Presenting my work at the last two ISME conferences, which greatly deepened my perspective on microbial ecology, are also worth-mentioning. A complete list with the conferences I have attended can be found on my cv.

Extensive IT skills, including experience with standard desktop software (e.g. office & graphics packages), research specific software (e.g. R stats) and online databases (e.g. European Nucleotide Archive, NCBI)

Among my tasks on the PREGO project was to harvest data from MGnify, an EMBL-EBI resource based on ENA, and the IMG/JGI. I have made use of all resources mentioned in the various projects described above.

Ability to meet deadlines Ability to contribute to broader management and administrative processes. Ability to assess and organise resources Excellent time management and organisational skills

While working on my research projects, during my PhD, I contributed as a leading writer to two successfully grant proposals:

1. an EOSC-Life proposal, which funded the last year of my PhD and led to the development of metagoflow (<https://imbbc.hcmr.gr/2021/05/22/>). This project was in collaboration with two research infrastructures, the European Marine Biological Research Centre (EMBRC) and the European Bioinformatics Institute(EMBL-EBI). My main contribution was the technical background, as well as the description of the main working packages regarding the development of the workflow. Dr. Pavloudi and I, being the main writers of the proposal, we managed to set the consortium, and submit the proposal in less than a three-weeks time. After Dr. Pavloudi left IMBBC, I became the main principal investigator of the project until my PhD was finished.

2. the CCMRI HFRI proposal of Dr. Pafilis, my PhD main supervisor where I was responsible for the "Text and data harvesting" work package.

On top of that, during my PhD, I helped organizing the policy for the HPC use, a really challenging task from which I learnt a lot on how assessing time and resources can be both experience, but also software-based.

After my PhD, I have found a way to both maintain my so-far software, and support the needs of both European Infrastructures that heavily use them, and keep up with the wide range of tasks at KU.

Excellent communication and interpersonal skills Proven ability to use initiative to efficiently plan, optimise and progress projects and communicate findings Proven ability to work as part of a collaborative team and to build and maintain effective working relationships with colleagues Ability and willingness to contribute to the work of others by offering practical and intellectual help Ability to work independently and as part of a team

Since my MSc thesis, I was introduced to a rather social way of doing science, where I would first discuss a lot with wet lab colleagues, who would explain to me the biological questions and the idiosyncrasies of their data, and then I would have to come up with a plan on how to deal with them from a bioinformatic perspective.

At the time, I was even involved in organizing Crete's second island sampling day. Along with Dr. Paragkamian and a great number of young researchers, we went through all the logistics and details to support this amazing experience of having more than 50 people across the island of Crete, sampling soil eDNA.

At KU and the Lab of Microbial Systems Biology, this was also the case. I had the chance to work side-by-side with a lot of wet lab people that generate data that will eventually be stored in the μ growthDB we have been developing. This was a great piece of team work, in terms of designing the database and its interface, based on the needs described by the people that would actually use it in the end, and then figuring out ways on how to implement it.

At a broader community level, together with Dr. Garza, we invest quite some time in building a core of metabolic modeling classes and how-to example cases

(https://github.com/hariszaf/metabolic_toy_model) . Just last week, we submitted a chapter on "*Sampling from the Solution Space and Metabolic Environments of Genome-Scale Metabolic Models*" to be published in an edition on "Flux Balance Analysis", in the protocol series *Methods in Molecular Biology*, by Springer Nature.

Last but not least, my commitment to open source projects, has helped me a lot to be in a position of contributing to the work of others.

Understand equal opportunity issues as they may impact on areas of research content.

Since this project addresses fundamental biology questions and given the fact that the data to be used come from volcanic hot springs, it is my belief that equal opportunity issues are less likely to affect the research content directly. Yet, I am mindful that collaboration and interpretation should remain open and inclusive, making sure that diverse perspectives are considered.

Desirable Skills, Knowledge and Experience

Experience building and analysing metabolic flux models

My contribution to the already described software, *microbetag* [7], *dingo* [11], and *dnngior* [12] highlight my experience in metabolic modeling. So does the recently published study of ours, where we evaluated interaction strength predictions, between pairs of species, using Flux Balance Analysis methods [6].

In addition, my ongoing projects, the *dingo-stats* Python library for the statistical analysis of the flux sampling data sets, and the *Salmonella infantis* and its probiotics, are also indicative of my experience with such models.

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Experience with a variety of different machine learning approaches

In the *dnngior* [12], the GEM gap-filling approach of ours, we trained a deep neural network on ≥ 11 k bacterial species to recover missing reactions.

In CellScanner we developed a framework combining neural networks and UMAP to distinguish counts from different microbial species in microbial community flow cytometry data using monoculture data to train our classifier. A manuscript on CellScanner is now in preparation.

Experience of directing and supervising practical computer and/or lab-based research of others (e.g. students, technicians)

Over the last years, I had the chance to supervise a number of students coming from different backgrounds, and exploring different topics for their bachelor or master thesis. I also discovered that I enjoy this a lot.

I had the chance to work with Ermis Delopoulos (<https://github.com/ermismd>), a master student in the bioinformatics program at KU Leuven. Neither of us had prior experience in Java, yet we needed to develop a Cytoscape application to make *microbetag* accessible in the most widely used network visualization platform in biology. Together, we developed the MGG app, and since some extra experience on Java would be more than useful, we submitted Ermis's thesis as a GSoC project, which was funded in 2023. He excelled in both GSoC and his master's thesis (defended with distinction), while I gained Java experience and advanced *microbetag* toward acceptance in the *Genome Biology* journal (*accepted in principle* since August 2025).

An even more challenging task was the supervision of a set of students over the last three years with whom we designed and developed the already mentioned μ growthDB. In the first year of my post-doc at KU Leuven, I had the chance to work with Julia Casado Gómez-Pallete (<https://github.com/jcasadogp>) a brilliant MSc student with an engineering background. With Julia we set the basic schema of this rather challenging resource that aims of organizing all the various experimental setups one may go for to get growth curve data. In the following year, Sofia Machado (<https://github.com/sofiamonsalve>) jumped in the project to develop a first version of both the back and the front-end, with the most challenging task being how to come up with a user-friendly way for describing the data to be submitted. Based on Sofia's draft, over the last year, an excellent programmer and keen in bioinformatics tasks Andrew Radev (<https://github.com/AndrewRadev/>), delivered a ready-to-use resource, that is now expecting users to submit their data and reach back to us for further improvements of the resource.

Sotiris Touliopoulos (<https://github.com/SotirisTouliopoulos/>) a bachelor student from Democritus University of Thrace, Greece, a member of the IGEM team of his university who had contacted me to help them with their project, decided to start a post-bachelor project with me. He managed to first enroll as a Google Summer of Code student in the Geom-Scale project, and then continue working with me doing his Erasmus+ traineeship at Prof's

Faust lab. Together, we have been investigating what one can do with the findings of flux sampling, and we have developed a Python package, called *dingo-stats*, for the statistical analysis of flux sampling results. We are now preparing a manuscript for publication.

A full list of the students I have supervised is available on my CV.

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